My Project

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Chapter 1

File Index

1.1 File List

Here is a list of all documented files with brief descriptions:

Chapter 2

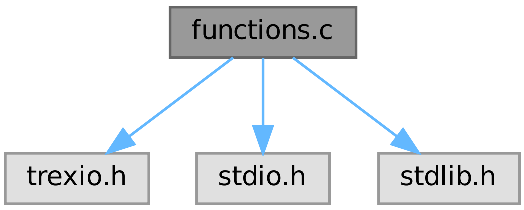
File Documentation

2.1 functions.c File Reference

Functions to read quantum chemical data using the TREXIO library.

```
#include <trexio.h>
#include <stdio.h>
#include <stdlib.h>
```

Include dependency graph for functions.c:



Functions

- trexio_t ∗ open_fun (const char ∗filename)

    Opens a TREXIO file.
- void Nuclear_repulsion_energy (trexio_t ∗trexio_file, double ∗Enn)

    Reads the nuclear repulsion energy.
- void Occupied_orbitals (trexio_t ∗trexio_file, int32_t ∗n_up)

    Reads the number of occupied orbitals.
- void Molecular_orbitals (trexio_t ∗trexio_file, int32_t ∗mo_num)

    Reads the total number of molecular orbitals.
- void ∗ one_e_integrals (trexio_t ∗trexio_file, int mo_num, double ∗∗data)

    Allocates memory and reads the one-electron integrals.
- void ∗ mo_energies (trexio_t ∗trexio_file, int mo_num, double ∗∗mo_energy)

Allocates memory and reads the molecular orbital energies.
- void number_2_e_integrals (trexio_t ∗trexio_file, int64_t ∗n_integrals)

  Reads the number of two-electron integrals.
- void two_e_integrals (trexio_t ∗trexio_file, int n_integrals, int32_t ∗∗index_out, double ∗∗value_out)

  Reads the indices (i, j, a, b) and the value of the two-electron integrals.
- void close_fun (trexio_t ∗trexio_file)

  Closes a TREXIO file.

2.1.1 Detailed Description

Functions to read quantum chemical data using the TREXIO library.

Author

David Gómez and Esther Lago

2.1.2 Function Documentation

2.1.2.1 close_fun()

```
void close_fun (
            trexio_t * trexio_file )
```

Closes a TREXIO file.

Parameters

trexio_file	Pointer to the TREXIO file structure to be closed.

2.1.2.2 mo_energies()

```
void * mo_energies (
            trexio_t * trexio_file,
            int mo_num,
            double ** mo_energy )
```

Allocates memory and reads the molecular orbital energies.

Reads the molecular orbital energies.

Parameters

trexio_file	Pointer to the open TREXIO file.
mo_num	Number of molecular orbitals (used for allocation size).
mo_energy	Pointer to a double pointer where the molecular orbital energies will be stored.

2.1.2.3 Molecular_orbitals()

```
void Molecular_orbitals (
            trexio_t * trexio_file,
            int32_t * mo_num )
```

Reads the total number of molecular orbitals.

Parameters

trexio_file	Pointer to the open TREXIO file.
mo_num	Pointer to an integer where the number of orbitals will be stored.

2.1.2.4 Nuclear_repulsion_energy()

```
void Nuclear_repulsion_energy (
            trexio_t * trexio_file,
            double * Enn )
```

Reads the nuclear repulsion energy.

Reads nuclear repulsion energy.

Parameters

trexio_file	Pointer to the open TREXIO file.
Enn	Pointer to a double where the energy value will be stored.

2.1.2.5 number_2_e_integrals()

```
void number_2_e_integrals (
            trexio_t * trexio_file,
            int64_t * n_integrals )
```

Reads the number of two-electron integrals.

Parameters

trexio_file	Pointer to the open TREXIO file.
n_integrals	Pointer to an int64 where the number of two-electron integrals will be stored.

2.1.2.6 Occupied_orbitals()

```
void Occupied_orbitals (
            trexio_t * trexio_file,
            int32_t * n_up )
```

Reads the number of occupied orbitals.

Parameters

trexio_file	Pointer to the open TREXIO file.
n_up	Pointer to an integer where the number of up-spin electrons will be stored.

2.1.2.7 one_e_integrals()

```
void * one_e_integrals (
            trexio_t * trexio_file,
            int mo_num,
            double ** data )
```

Allocates memory and reads the one-electron integrals.

Reads one-electron integrals.

Parameters

trexio_file	Pointer to the open TREXIO file.
mo_num	Number of molecular orbitals (occupied + virtuals).
data	Pointer to a double pointer where the one-electron integrals will be stored.

2.1.2.8 open_fun()

```
trexio_t * open_fun (
            const char * filename )
```

Opens a TREXIO file.

Parameters

filename	String containing the path to the TREXIO file (HDF5 format).

Returns

trexio_t∗ Pointer to the TREXIO file structure. Exits on error.

2.1.2.9 two_e_integrals()

```
void two_e_integrals (
            trexio_t * trexio_file,
            int n_integrals,
            int32_t ** index_out,
            double ** value_out )
```

Reads the indices (i, j, a, b) and the value of the two-electron integrals.

Parameters

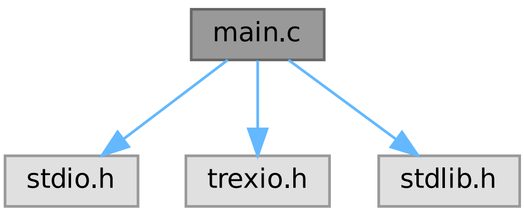
trexio_file	Pointer to the open TREXIO file.
n_integrals	Number of two-electron integrals.
index	Pointer to an int32 pointer where the indices of the two-electron integrals will be stored.
value	Pointer to a double pointer where the value of the two-electron integrals will be stored.

2.2 main.c File Reference

Program to read data from a TREXIO file using it to compute the Hartree-Fock and MP2 energies.

```
#include <stdio.h>
#include <trexio.h>
#include <stdlib.h>
```

Include dependency graph for main.c:



Macros

- #define pointer(i, j, k, l) ((i)∗mo_num∗mo_num∗mo_num + (j)∗mo_num∗mo_num + (k)∗mo_num +(l))

  Macro to map 4D orbital indices to a linear 1D index.

Functions

- trexio_t ∗ open_fun (const char ∗filename)

  Opens a TREXIO file.
- void Nuclear_repulsion_energy (trexio_t ∗trexio_file, double ∗Enn)

  Reads nuclear repulsion energy.
- void Occupied_orbitals (trexio_t ∗trexio_file, int32_t ∗n_up)

  Reads the number of occupied orbitals.
- void Molecular_orbitals (trexio_t ∗trexio_file, int32_t ∗mo_num)

  Reads the total number of molecular orbitals.
- void ∗ one_e_integrals (trexio_t ∗trexio_file, int mo_num, double ∗∗data)

  Reads one-electron integrals.
- void ∗ mo_energies (trexio_t ∗trexio_file, int mo_num, double ∗∗mo_energy)

Reads the molecular orbital energies.
- void number_2_e_integrals (trexio_t *trexio_file, int64_t *n_integrals)

  Reads the number of two-electron integrals.
- void * two_e_integrals (trexio_t *trexio_file, int n_integrals, int32_t **index, double **value)

  Reads the indices (i, j, a, b) and the value of the two-electron integrals.
- void close_fun (trexio_t *trexio_file)

  Closes a TREXIO file.
- double **main** ()

2.2.1 Detailed Description

Program to read data from a TREXIO file using it to compute the Hartree-Fock and MP2 energies.

Authors

David Gómez and Esther Lago

2.2.2 Macro Definition Documentation

2.2.2.1 pointer

```
#define pointer(
            i,
            j,
            k,
            l ) ((i)*mo_num*mo_num*mo_num + (j)*mo_num*mo_num + (k)*mo_num +(l))
```

Macro to map 4D orbital indices to a linear 1D index.

Parameters

i,j,k,l	Orbital indices.

2.2.3 Function Documentation

2.2.3.1 close_fun()

```
void close_fun (
            trexio_t * trexio_file )
```

Closes a TREXIO file.

Parameters

trexio_file	Pointer to the TREXIO file structure to be closed.

2.2.3.2 mo_energies()

```
void * mo_energies (
            trexio_t * trexio_file,
            int mo_num,
            double ** mo_energy )
```

Reads the molecular orbital energies.

Reads the molecular orbital energies.

Parameters

trexio_file	Pointer to the open TREXIO file.
mo_num	Number of molecular orbitals (used for allocation size).
mo_energy	Pointer to a double pointer where the molecular orbital energies will be stored.

2.2.3.3 Molecular_orbitals()

```
void Molecular_orbitals (
            trexio_t * trexio_file,
            int32_t * mo_num )
```

Reads the total number of molecular orbitals.

Parameters

trexio_file	Pointer to the open TREXIO file.
mo_num	Pointer to an integer where the number of orbitals will be stored.

2.2.3.4 Nuclear_repulsion_energy()

```
void Nuclear_repulsion_energy (
            trexio_t * trexio_file,
            double * Enn )
```

Reads nuclear repulsion energy.

Reads nuclear repulsion energy.

Parameters

trexio_file	Pointer to the open TREXIO file.
Enn	Pointer to a double where the energy value will be stored.

2.2.3.5 number_2_e_integrals()

```
void number_2_e_integrals (
```

```
            trexio_t * trexio_file,
            int64_t * n_integrals )
```

Reads the number of two-electron integrals.

Parameters

trexio_file	Pointer to the open TREXIO file.
n_integrals	Pointer to an int64 where the number of two-electron integrals will be stored.

2.2.3.6 Occupied_orbitals()

```
void Occupied_orbitals (
            trexio_t * trexio_file,
            int32_t * n_up )
```

Reads the number of occupied orbitals.

Parameters

trexio_file	Pointer to the open TREXIO file.
n_up	Pointer to an integer where the number of up-spin electrons will be stored.

2.2.3.7 one_e_integrals()

```
void * one_e_integrals (
            trexio_t * trexio_file,
            int mo_num,
            double ** data )
```

Reads one-electron integrals.

Reads one-electron integrals.

Parameters

trexio_file	Pointer to the open TREXIO file.
mo_num	Number of molecular orbitals (occupied + virtuals).
data	Pointer to a double pointer where the one-electron integrals will be stored.

2.2.3.8 open_fun()

```
trexio_t * open_fun (
            const char * filename )
```

Opens a TREXIO file.

Parameters

filename	String containing the path to the TREXIO file (HDF5 format).

Returns

trexio_t∗ Pointer to the TREXIO file structure. Exits on error.

2.2.3.9 two_e_integrals()

```
void * two_e_integrals (
            trexio_t * trexio_file,
            int n_integrals,
            int32_t ** index_out,
            double ** value_out )
```

Reads the indices (i, j, a, b) and the value of the two-electron integrals.

Parameters

trexio_file	Pointer to the open TREXIO file.
n_integrals	Number of two-electron integrals.
index	Pointer to an int32 pointer where the indices of the two-electron integrals will be stored.
value	Pointer to a double pointer where the value of the two-electron integrals will be stored.